University College London

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PHAR0002: An introduction to the mechanisms of drug action.

Full agonists, partial agonists and constitutively active receptors.

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This handout has been modified from the original text developed by Prof. D.H. Jenkinson who taught this course for many years and eventually converted his notes into an excellent chapter in the *Textbook of Receptor Pharmacology*, Ed., Foreman, J.C., and Johansen, T., CRC Press, 1996/2003/2010.

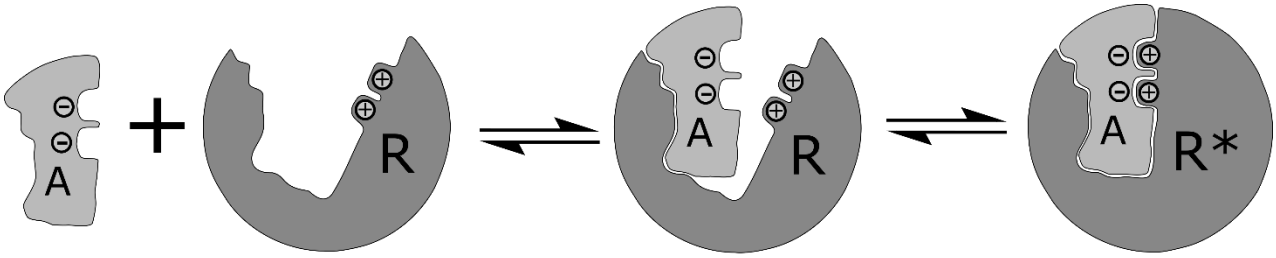
Aims:

1) To know what is meant by the terms 'full agonist' and 'partial agonist'.
2) To understand the dose-response curve seen with a partial agonist alone, or when a partial agonist is used in the presence of a full agonist.
3) To understand the idea of agonist efficacy.
4) To be able to explain the activity of a partial agonist at the molecular level and to understand the difference between receptor binding and receptor activation.
5) To know what is meant by a constitutively active receptor.

Why receptors activate when agonist is applied.

You should, by now, be used to the idea that receptors (such as G-protein-coupled receptors or ligand-gated ion channels) are activated by the application of agonist. A very fundamental question you might ask yourself is 'why? – what does the agonist do?'. To start to understand the answer to this question we must think about the physical picture of agonist and receptor. Taking the example of a ligand-gated ion channel, we know that activation means the channel opens. Clearly, for this to happen there has to be a conformational change in the receptor. Indeed, all agonist activations produce conformational changes in the receptor molecule. So, now the question is 'why does a conformational change occur?'. The answer is this; on binding an agonist molecule the receptor is able to make bonds (for example, hydrogen bonds or charge-charge interactions etc) that favour the receptor occupying the active conformation. A very simplified way to think about this is illustrated in Figure 1. In this model we imagine that negative charges of an agonist molecule interact with positive charges on the receptor and favour the active receptor conformation, which, to be consistent with the schemes below, is marked 'R*', rather than R so as to indicate that the receptor is active. (Although in this illustration we have indicated charge-charge interactions between receptor and agonist, this is often not the type of bond formed and hydrogen bonds or hydrophobic interactions are just as likely.) Notice that this conformational change is reversible; at some point the receptor-agonist

complex will have enough thermal energy to break the bonds that hold them in the active conformation.



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Figure 1. A cartoon of agonist action. In this situation we imagine a receptor with positive charges somewhere on the protein, and an agonist molecule, with negative charges. When the agonist binds, negative charges on the agonist attract the positive charges on the receptor and hence favour the adoption of an active conformational state denoted 'R*'.

In an earlier lecture we examined an approach to understanding the dose-response curve in which the agonist interacts with its receptor according to the following scheme:

$$A + R \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} AR$$

We can now see that this scheme is inadequate to represent the above picture because it does not differentiate between the R and R* states (Figure 1). It would be more appropriate to use a scheme of the following kind:

$$A + R \overset{K_1}{\rightleftharpoons} AR \overset{E}{\rightleftharpoons} AR^*$$

In this revised model, the first step, with equilibrium constant K_1, relates to the binding of agonist and the second step, with equilibrium constant E, represents the conformational change to an active state (the gating step) and the ratio of the forward to the backward rates. This scheme is the ***del Castillo-Katz model of receptor activation*** and was the first model to separate agonist binding and receptor gating steps in the pathway towards receptor activation. The separate step for activation embodies the idea of agonist efficacy – the ability of the agonist to stabilise the active state of the receptor. We will return to this model shortly. However, for the moment we will take a small detour and consider a second, but closely related the topic; 'partial agonists'.

PARTIAL AGONISTS, FULL AGONISTS AND EFFICACY

The development of new drugs commonly requires the synthesis of a large number of structurally related compounds. If agonists of this kind are tested on a particular tissue, they are often found to fall into two groups. Some are able to elicit a maximal tissue response, and accordingly are described as ***full agonists***. The others cannot elicit this maximal response, no matter how high their concentration, and are termed ***partial agonists***. Examples include

Partial agonist	**Full agonist**	**Acting at:**
prenalterol	adrenaline, isoprenaline	β-adrenoceptors
impromidine	histamine	histamine H_2 receptors

Figure 2 shows concentration response curves that compare the action of the □-adrenoceptor partial agonist prenalterol with that of the full agonist isoprenaline on a range of tissues and responses. In every instance, the maximum response to prenalterol is smaller, though the magnitude of the difference varies greatly.

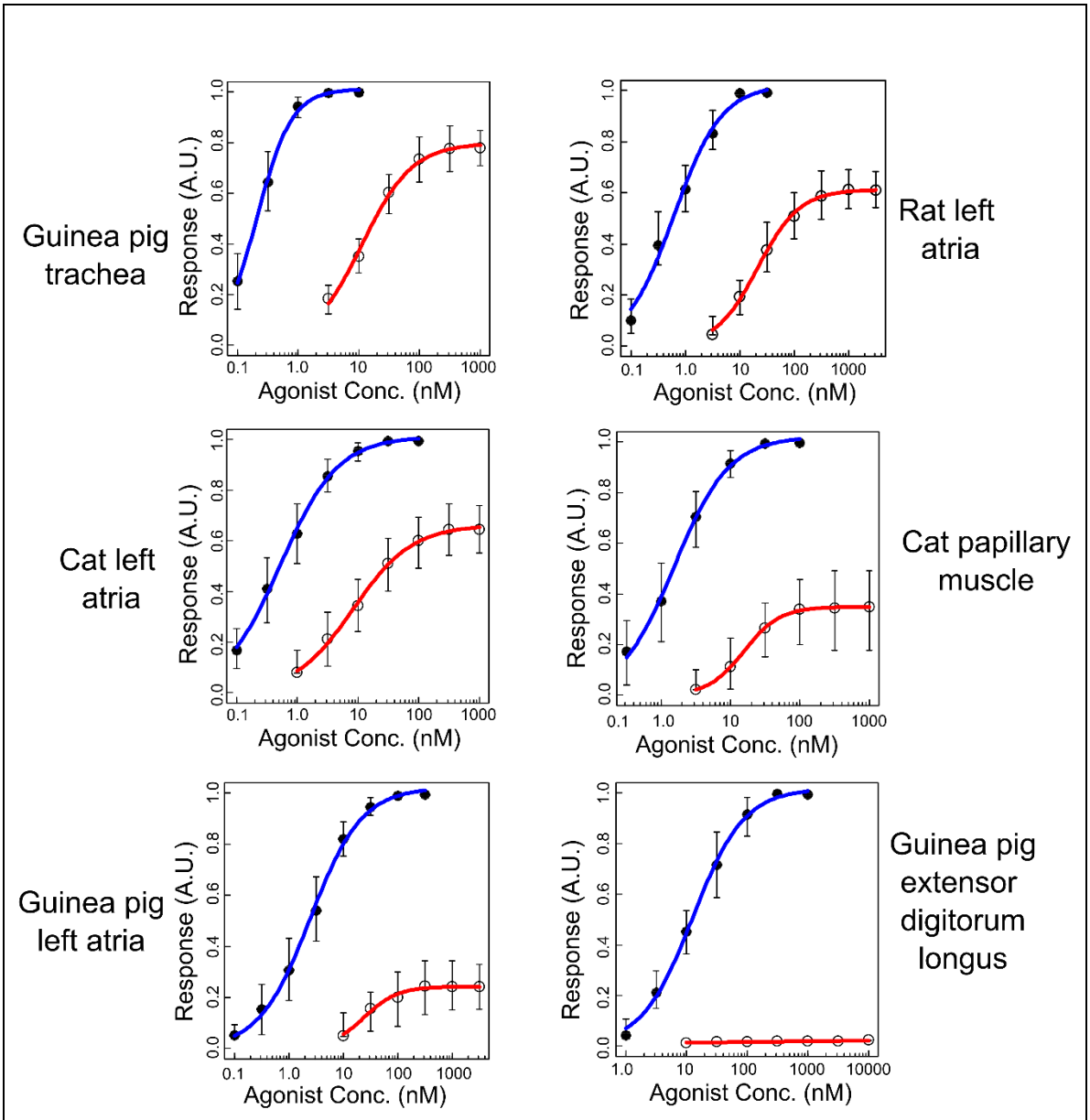


Figure 2. Comparison of the log concentration-response relationships for β-adrenoceptor mediated actions, on six tissues, of a full and a partial agonist (isoprenaline blue (●) and prenalterol red (○) respectively). The ordinate shows the response as a fraction of the maximal response to isoprenaline. (Kenakin, T. P. and Beek, D., *J. Pharmacol. exp. Ther*. 213, 406 – 413, 1980)

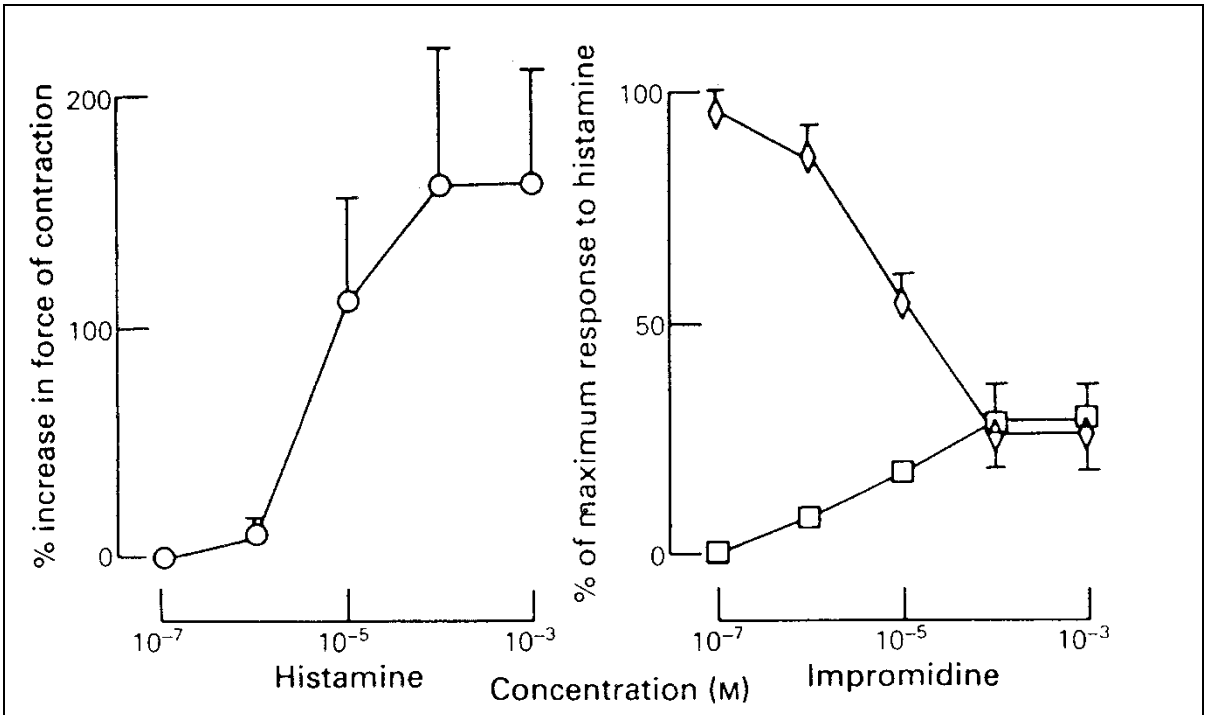


Figure 3. Interaction between histamine and the H_2-receptor partial agonist impromidine on isolated ventricular strips from human myocardium. The log concentration-response curve on the left shows the response to histamine alone, and those on the right show responses to impromidine acting either alone (□) or in the presence of a constant concentration (10^{-4} M) of the full agonist histamine (◊). From T.A.H. English *et al*., *Br. J. Pharmacol.*, **89**, 335 - 340 (1986).

A superficial explanation for the failure of a partial agonist to elicit as large a response as a full agonist could be that it is not able to combine with all the receptors. This can easily be ruled out by testing the effect of increasing concentrations of a partial agonist on the response of a tissue to a fixed concentration of a full agonist. Figure 3 illustrates such an experiment. As the concentration of the partial agonist is raised, the response of the tissue gradually falls from the large value seen with the full agonist alone, and eventually reaches the maximum response to the partial agonist acting on its own. The implication is that the partial agonist is perfectly able to combine with all the receptors, provided that a high enough concentration is applied, but that the tissue response that results is less than would be seen with a full agonist. The partial agonist is in some way less able to activate the receptors.

The experiment of Figure 4 points to the same conclusion. When low concentrations of histamine are applied in the presence of the partial agonist impromidine, the response is larger than that to impromidine alone. However, the concentration-response curves cross as the histamine concentration is further increased. This is because the presence of impromidine reduces receptor

occupancy by histamine (at all concentrations). When the lines intersect, the reduction in the contribution of histamine to the response is no longer fully offset by the contribution from the receptors occupied by impromidine. Beyond this point, the presence of impromidine reduces the response to histamine. Again, the implication is that the partial agonist is capable of combining with all the receptors, but the response is less than with a full agonist.

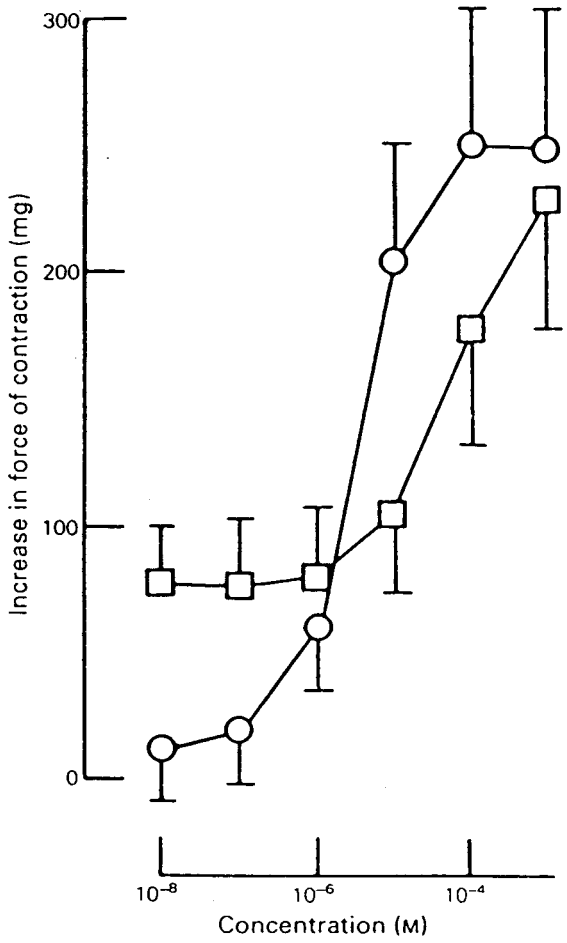


Figure 4. Log concentration - response curves for histamine applied alone (○) or in the presence (□) of a constant concentration of the partial agonist impromidine (10^{-5} M). Tissue and experimental conditions as in Figure 3. From T.A.H. English *et al.*, *Br. J. Pharmacol.,* **89**, 335 - 340 (1986).

MODELS TO EXPLAIN PARTIAL AGONISM

This apparently strange behaviour of partial agonists can be explained by extending the model in Figure 1. In that model, the agonist was a molecule that made it energetically more favourable for the receptor to be in an active state through favourable charge-charge interactions that stabilise the active conformation. This might be like having a supportive chair with arm rests and good back support to make it easier for you to assume an alert 'active' conformation in my lectures. To extend this idea, just as some chairs will provide better support in the active conformation than others, some agonists are better able to stabilise the active receptor conformation than others. This is not

because they bind more tightly but because the bonds and interactions they make with the receptor are better able to stabilise the active conformation. Thus, the crucial difference between full and partial agonists would be apparent in the second half of the del Castillo-Katz mechanism and we might thus expand Figure 1 as shown in Figure 5.

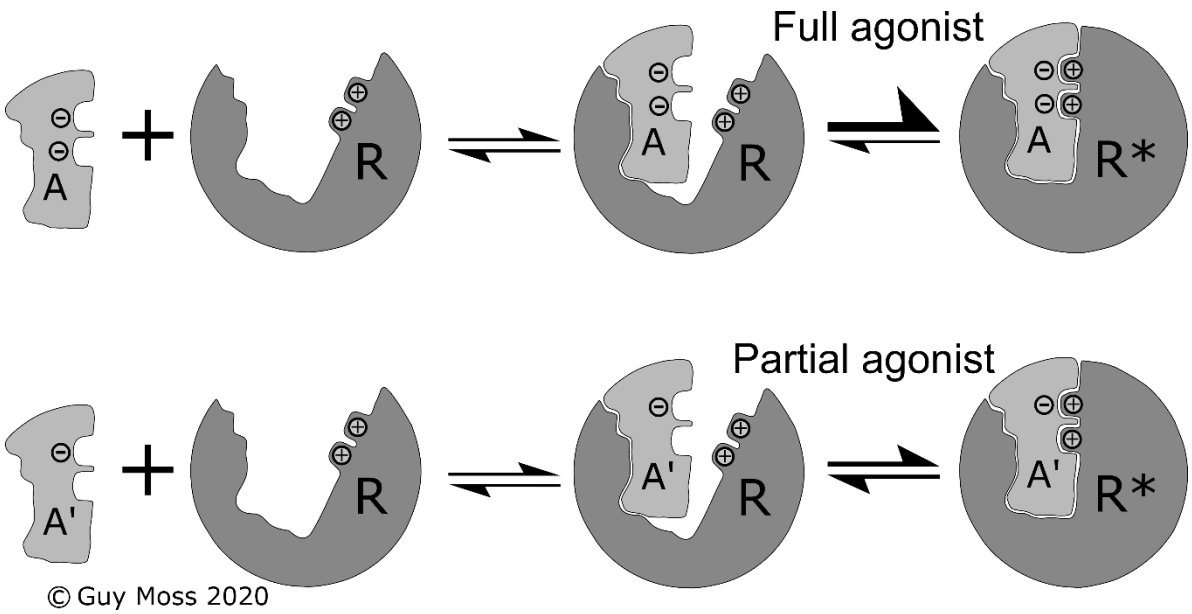


Figure 5. A cartoon illustrating full and partial agonists. The crucial difference between a partial agonist and a full agonist lies in their ability to activate (rather than bind to) the receptor. Extending the ideas of Figure 1, we might imagine that a partial agonist is less able to stabilise the receptor in its active state because, in our imagined scheme, it has one less negative charge. With this change, the forward transition rate to an active conformation may be slow whilst the rate constant for the transition from AR* → AR will be faster because there are fewer bonds holding things in the active state. Arrows of different sizes they have been drawn to emphasise this point.

<u>Confirmation of the del Castillo-Katz model.</u>

In 1976, E. Neher and B. Sakmann devised a method for recording the minute current which flows when a single ligand-gated ion channel receptor is activated by agonist (the single channel recording method). Applying this method to study the nicotinic receptor it was quickly shown that for a wide range of nicotinic agonists, the current flowing through the receptor was always of the same size. What differed between the agonists was the proportion of time for which the current flowed, i.e., for which the channels were open. Further, as might be expected, in these single channel records it is possible to see the channel close for both long periods (between agonist binding events) and brief periods (when the receptor 'hops' between active, AR* -like, and inactive AR-like states. The records thus provided a confirmation of the principle of separate gating and binding steps laid down in the del-Castillo and Katz model. The interpretation of such records in terms of states R, AR and AR* is illustrated in Figure 6.

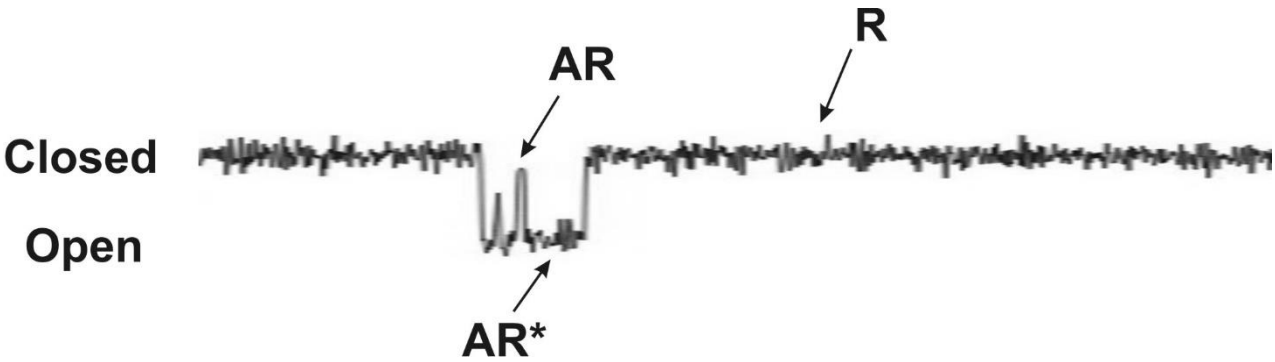


Figure 6. The current recorded from a single nicotinic-type acetylcholine receptor ion channel. Most of the time this skeletal muscle channel is closed, waiting for an agonist molecule to arrive. When agonist and receptor do form a complex the channel opens in a 'burst' of activity. This behavior can be explained (qualitatively) by our partial agonist model. The long shut times are periods where the channel is waiting to associate with another agonist molecule. The brief shut times, however, are surrounded by openings and this 'burst' of activity probably corresponds to rapid transitions between the AR and the AR* state during which time the ligand is always bound. Note: although this is fine as a qualitative explanation, detailed analysis of these channels shows that more complicated schemes are necessary to fully explain the data. (Recording courtesy of Dr. Chris Shelley UCL).

The slow-channel congenital myasthenic syndromes

The importance of both gating and binding is well illustrated by a very rare form of inherited genetic disorder called slow channel congenital myasthenia. The slow-channel congenital myasthenic syndromes (SCCMS) are disorders of neuromuscular transmission. They are characterized by muscle weakness and fatigability. Although a variety of mutations cause SCCMS, all SCCMS mutations produce prolonged endplate currents. At the single channel level this is evident in changes of both the binding and efficacy of acetylcholine. These changes are often very striking as illustrated in Figure 7. One consequence of these primary defects in acetylcholine receptor is a degeneration of the junctional folds and this probably accounts for the muscle weakness.

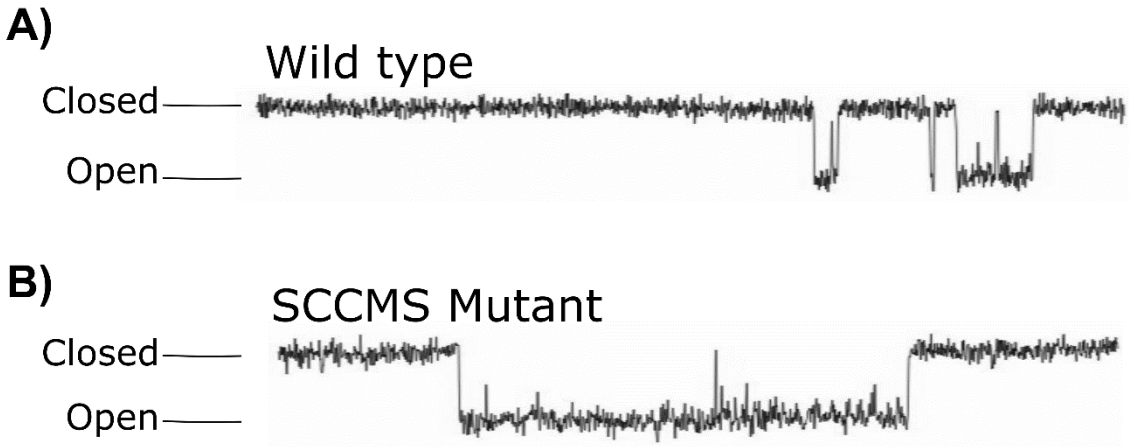


Figure 7. Two single channel records of acetylcholine-activated ion channels, illustrating the long burst behaviour of SCCMS mutant channels. A) The current flowing through a single nicotinic receptor ion channel showing the behavior seen in healthy 'wild type' channels. The channel is mostly closed but opens in little bursts (downwards steps in the record) when acetylcholine is bound. Channel openings are shown downwards, as is the convention for showing currents flowing into a cell. B) The response of a channel carrying mutation epsilon L78P, a mutation responsible for some cases of slow channel congenital myasthenia. Notice the much longer open time of this channel - detailed analysis reveals that such mutants differ in both acetylcholine efficacy and binding. (Recordings courtesy of Dr. Chris Shelley UCL, Pharmacology).

Crystal structures of full and partial adrenoceptor agonists.

In 2011, a UK-based team determined the structure of β1-adrenoceptors (one of the G-protein couple receptors we discussed in earlier lectures) with both full and partial agonists bound (Warne et al., 2011, Nature 469, 241). They were able to show that both full and partial agonists encourage the same conformational change in the receptor. However, a full agonist was able to stabilize that conformation by making two hydrogen bonds with serine residues 211 and 215 of the receptor, while partial agonists could only make one of these interactions with serine 211 (Figure 8).

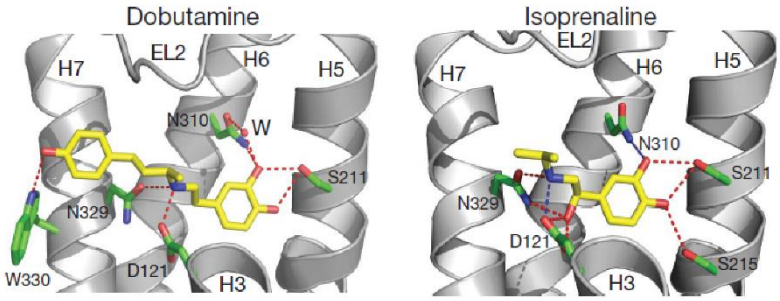


Figure 8. The binding of partial agonist dobutamine utilizes a hydrogen bond with serine 211, while full agonist isoprenaline hydrogen bonds with both serine 211 (S211) and serine 215 (S215). Image from Warne et al 2011.

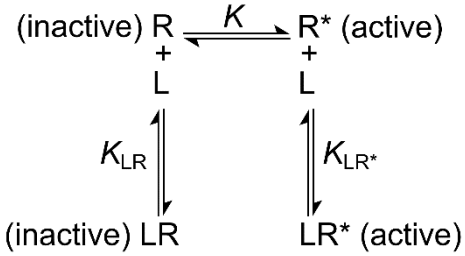
These ideas thus support the simple, del Castillo-Katz model. However, there is also good evidence that more complex models apply in some cases where different agonists producing distinct active conformations of GPCRs, as will be discussed in later years.

CONSTITUTIVELY ACTIVE RECEPTORS

In thinking about receptor activation, we have created a picture in which receptors do not assume an active conformation in the absence of agonist because too much energy is required; they need the 'help' of agonist-receptor bonds. In fact, for many cases this is not quite true. The energy required to reach the active state without agonist simply makes it very unlikely that a receptor will assume the correct conformation. However, sometimes, by chance, random collisions can provide the thermal energy that is required to enter the active state. When receptors activate in the absence of agonist, they are said to be *constitutively active*.
We might replace our earlier simple agonist/receptor scheme with a more complicated system in the absence of agonist.

$$\underset{\text{(inactive)}}{R} \rightleftharpoons \underset{\text{(active)}}{R^*}$$

In principle, both forms could combine with agonist, or indeed with any ligand, *L*, with affinity:



Suppose that *L* combines only with the inactive, R, form. Then the presence of *L*, by promoting the formation of *L*R at the expense of the other species, will *reduce* the proportion of receptors in the active, R*, state, and L is said to be an ***inverse agonist*** or ***negative antagonist.*** If, in contrast, *L* combines with the R* form alone, the outcome will be an increase in the proportion in the active state, and so *L* will behave as a conventional agonist. If, however, *L* has equal affinity for R and R*, its presence will not affect the fraction of receptors in the active state. Such a ligand is called a reversible competitive antagonist and we will be examining their properties in the next lecture. Overall, our picture of agonist receptor interactions, outlined in Figures 1 and 5 could be extended further still as shown in Figure 9.

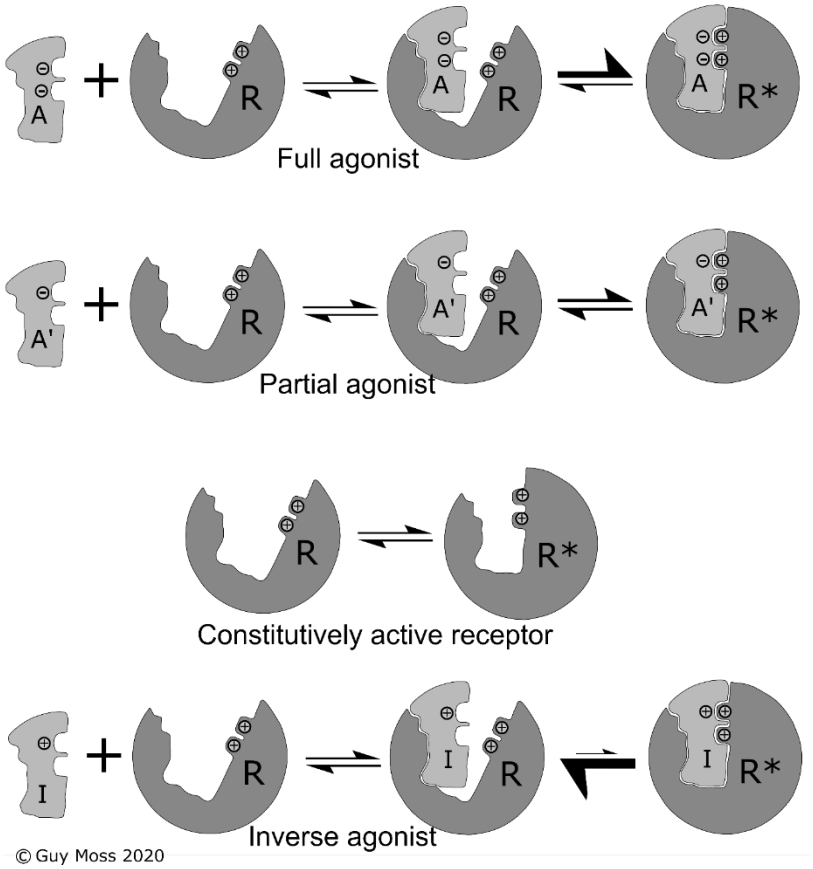


Figure 9. An extension to the picture of agonist activation to include inverse agonists, and the idea of constitutively active receptors. To continue the simple idea shown in figure 1, we can imagine an inverse agonist as a molecule carrying positive charge instead of negative charge. Again, the relative sizes of the arrows in each equilibrium are designed to illustrate, qualitatively, how the rate constants associated with receptor activation might change.

DESENSITIZATION

A complication likely to be met in any detailed study of the relationship between the concentration of an agonist and the action it elicits, is the occurrence of ***desensitization***. The response declines despite the continued presence of the agonist. Several factors can contribute. One that has been established in work with 'fast' receptors (ligand gated ion channels) is that receptors occupied by agonist, and in the active state (AR*), may isomerise to an inactive, desensitized, state, AR^*_D :

$$\underset{\text{(inactive)}}{A + R} \overset{K_1}{\rightleftharpoons} \underset{\text{(inactive)}}{AR} \overset{K_2}{\rightleftharpoons} \underset{\text{(active)}}{AR^*} \overset{K_3}{\rightleftharpoons} \underset{\text{(inactive)}}{AR^*_D}$$

Desensitization can occur in other ways. With G protein-coupled receptors, it can result from phosphorylation of the receptor by one or more protein kinases which become active following the application of agonist. This may be followed by the loss of some of the receptors from the cell surface.

SPARE RECEPTORS.

We have provided an explanation for partial agonists, but can we explain the variation in maximal response seen when we test these partial agonists on various tissues (Figure 2)? Surely we would expect the partial agonist to always produce the same maximum response? One solution would be to suggest that there are different receptor subtypes in these different tissues and that the agonist efficacy varies with the receptor subtype. This is entirely possible. However, frequently the issue is one of **'spare receptors'**.

Tissues in the human body generally have more receptors than are needed to produce a maximum agonist response. For example, maximal muscle contraction can still occur when a significant fraction of the ACh receptors are blocked at the neuromuscular junction. Similarly, the action potential is still propagated even when a significant proportion of the sodium channels are blocked. The reason for this is that in any coupling of receptors to a tissue response there will be a step that can saturate. For example, if sufficient ion channels open to fully depolarize a cell, opening more will not give a bigger response. The fact that tissues contain more receptors and are needed to produce a maximal response has led to the idea of 'spare receptors'. This does not, of course, mean that there are two kinds, 'spare' and 'used' - the receptors do not differ. However, only a few need to be activated to cause a large, or even maximal, response. This can occur when the response of the tissue is limited not by the receptors but by some subsequent event occurring as a consequence of receptor activation. For example, a piece of smooth muscle may not be able to shorten further. This maximal shortening may occur in response to a rise in cytosolic calcium that is much less than can be elicited by activating all the receptors. Thus, with a partial agonist, the number of spare receptors in a tissue plays a big role in determining the maximum response. Indeed, a partial agonist for one tissue could be a full agonist, at the same receptors, on another tissue if there is a big enough receptor reserve!